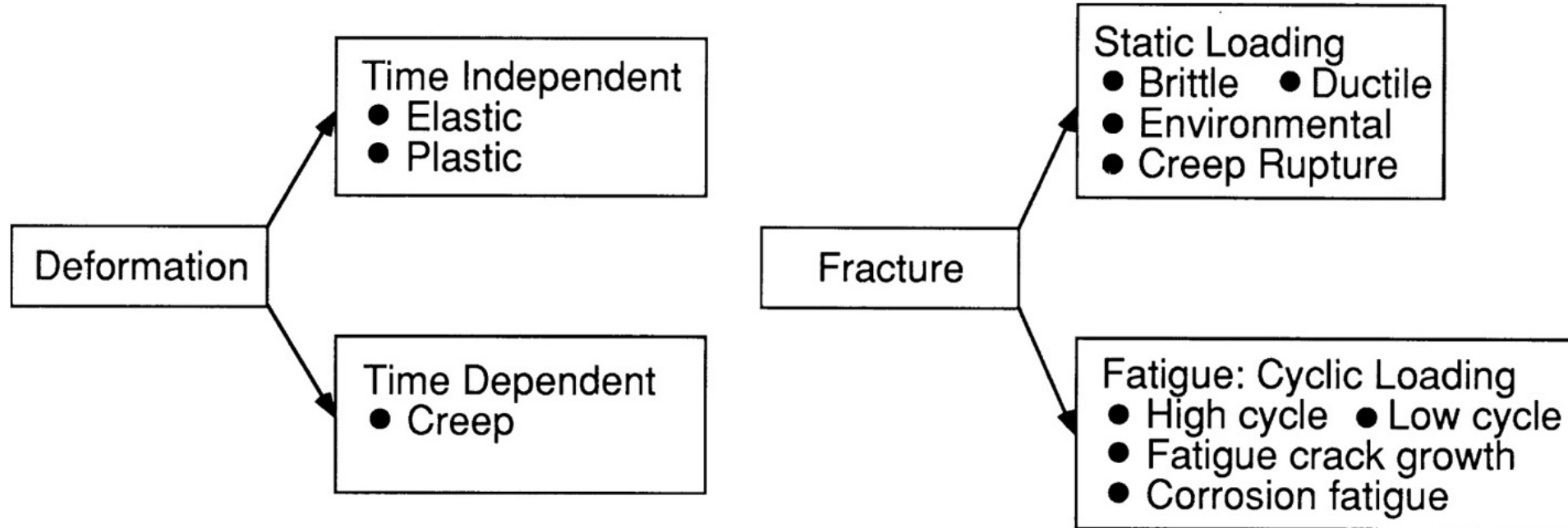


CH 6 & 7 Properties of Solid Materials

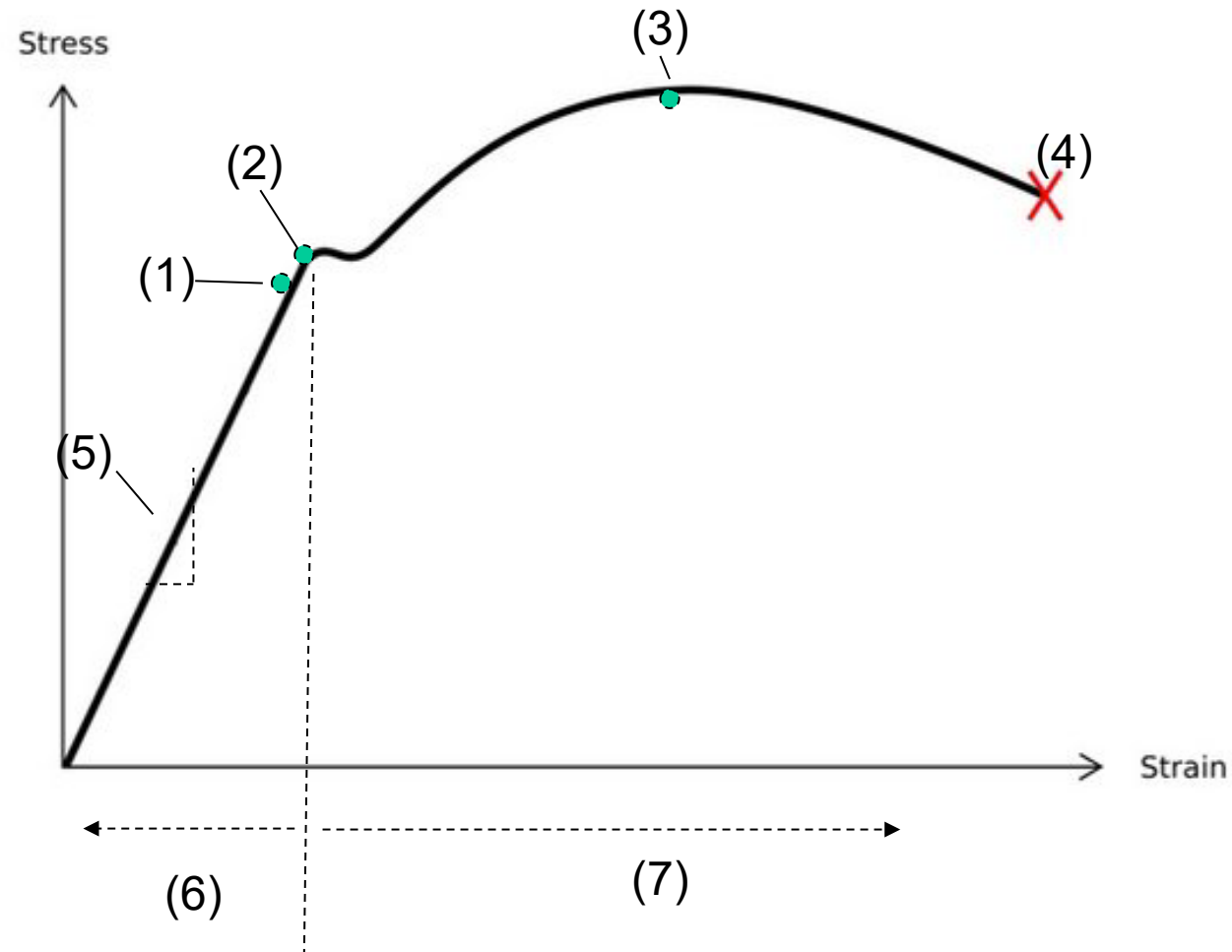
Mechanical Behavior of Materials



Upcoming Due Dates:

- Today: HW5 Due
- Module 2 Video – Sunday by 11:59 PM

Review:



1. Proportional Limit
2. Yield Stress
3. Ultimate (Tensile) Stress
4. Fracture Stress
5. Modulus of Elasticity
6. Elastic Behavior
7. Plastic Behavior

Review:

Why do engineers use **stress** and **strain** rather than **force** and **displacement**?

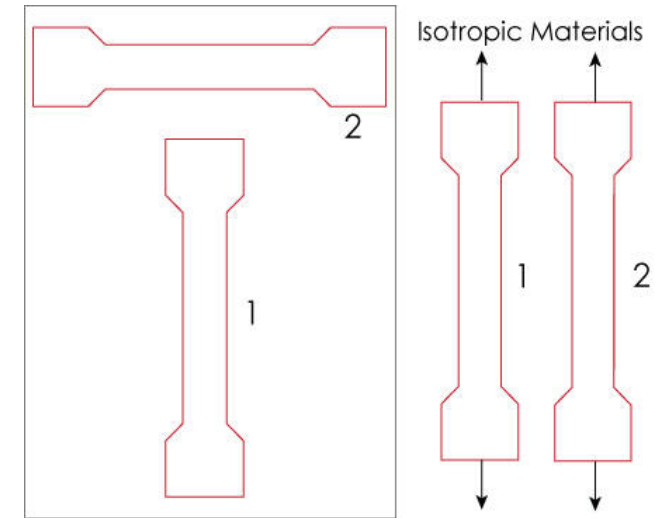
Stress (Strength) and Strain (% elongation) are in properties independent of geometry.

Types of Materials

Isotropic vs. Anisotropic

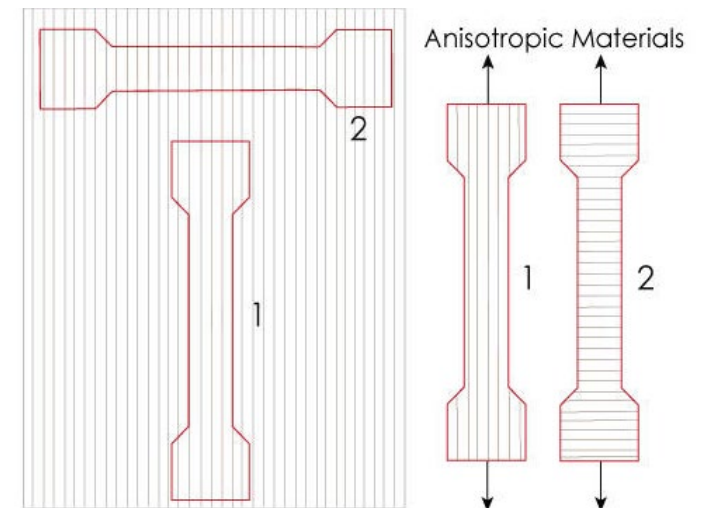
Isotropic:

- Properties are the same for all directions.
 - grains randomly oriented polycrystals
 - Ex: Steel, Aluminum



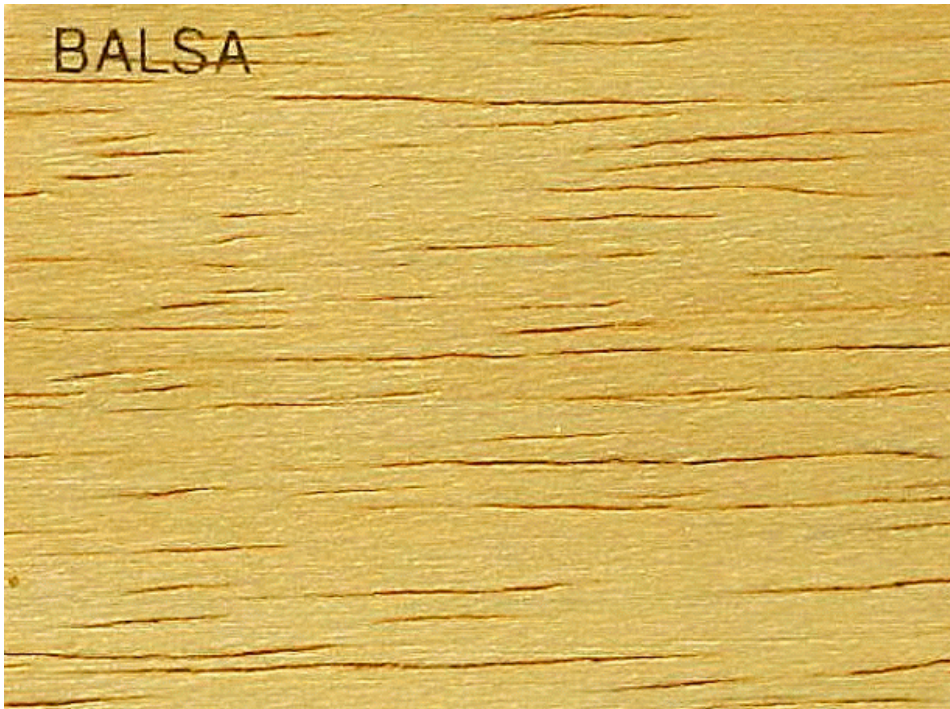
Anisotropic:

- Properties vary with directions.
 - single crystal; grains textured polycrystals
 - Ex: Wire reinforced concrete, biological materials



Biological Tissues

Most are anisotropic



Anisotropic:
Grains are aligned in same direction

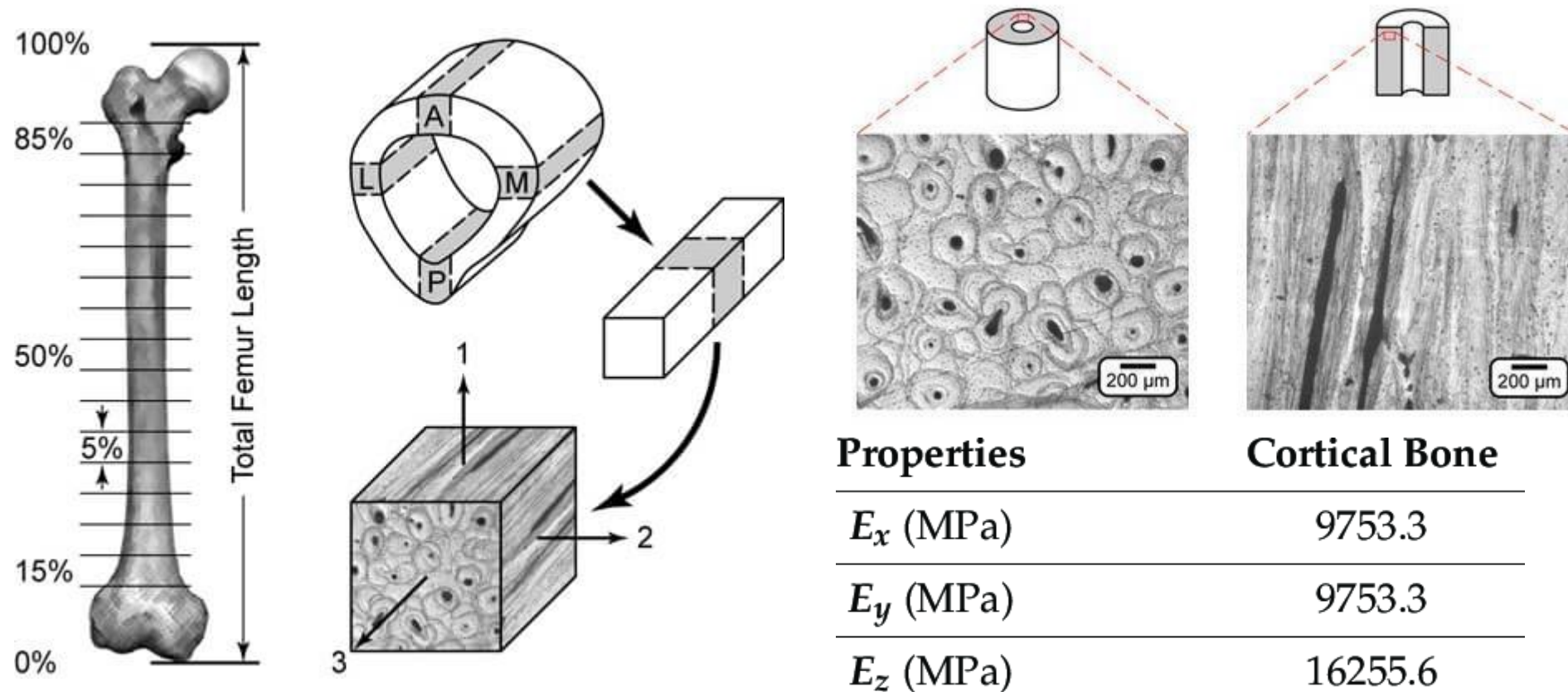
Vs.



Isotropic:
Same in all directions

Bone: Orthotropic Material

Orthotropic Materials: A type of anisotropy where Material properties differ along three **orthographic** axes



Crystals as Building Blocks

Some engineering applications require single crystals:

-- diamond single crystals for abrasives



(Courtesy Martin Deakins,
GE Superabrasives, Worthington, OH. Used
with permission.)

-- turbine blades



Properties of crystalline materials often related to crystal structure.

Ex: Quartz fractures more easily along some crystal planes than others.



(Courtesy P.M. Anderson)

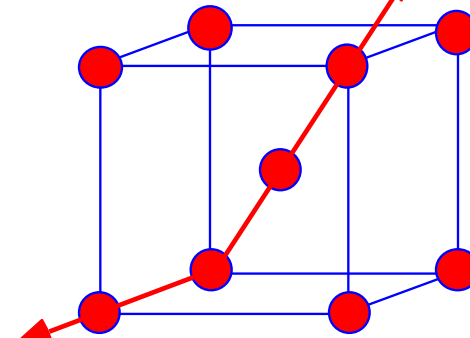
Fig. 9.40(c), *Callister & Rethwisch 3e*. (Fig. 9.40(c) courtesy of Pratt and Whitney).

Single vs Polycrystals

Single Crystals

- Properties vary with direction: **anisotropic**.
- Example: the modulus of elasticity (E) in BCC iron:

E (diagonal) = 273 GPa



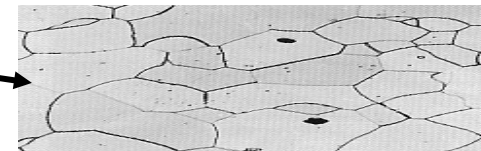
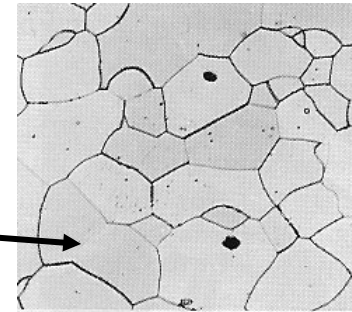
E (edge) = 125 GPa

Data from Table 3.7, *Callister & Rethwisch 3e*. (Source of data is R.W. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, 3rd ed., John Wiley and Sons, 1989.)

Polycrystals

- Properties may/may not vary with direction.
- If grains are randomly oriented: isotropic.
($E_{\text{poly iron}} = 210 \text{ GPa}$)
- If grains are textured, anisotropic.

200 μm



Adapted from Fig. 5.19(b), *Callister & Rethwisch 3e*. (Fig. 5.19(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Most Engineering Materials are Polycrystals



Adapted from Fig. K, color inset pages of *Callister 5e*.

(Fig. K is courtesy of Paul E. Danielson, Teledyne Wah Chang Albany)

Anisotropic

Isotropic

- Niobium, hafnium, and tungsten (Nb-Hf-W) plate with an electron beam weld.
- Each "grain" is a single crystal.
- If grains are randomly oriented, overall component properties are not directional.
- Grain sizes typ. range from 1 nm to 2 cm
 - (i.e., from a few to millions of atomic layers).

Other Elastic Properties

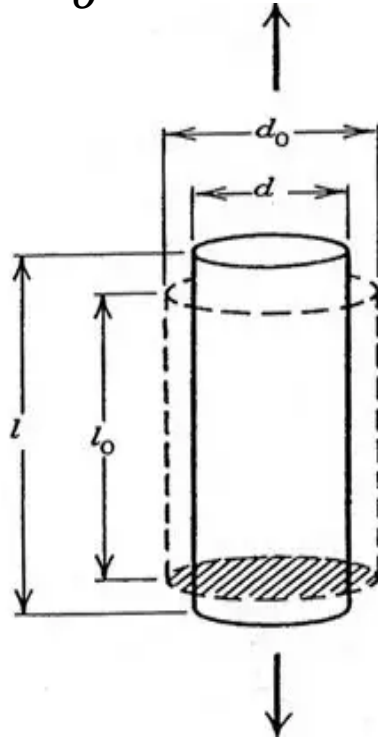
Engineering Strain

Longitudinal strain:

$$\epsilon_{long} = \frac{\Delta l}{l_o} = \frac{l - l_o}{l_o}$$

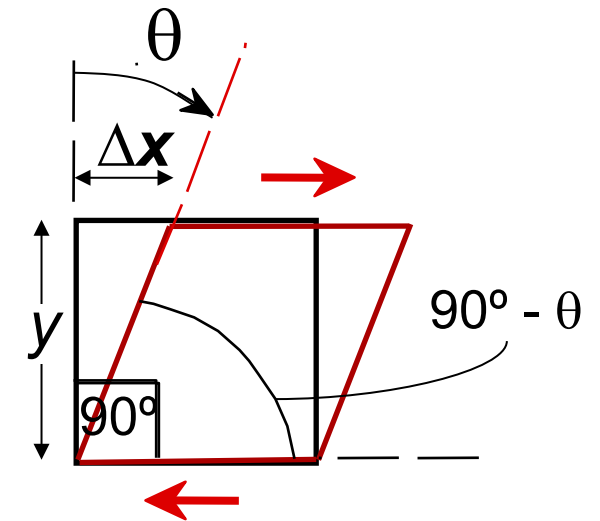
Lateral strain:

$$\epsilon_{lat} = \frac{\Delta d}{d_o} = \frac{d - d_o}{d_o}$$



Shear strain:

$$\gamma = \Delta x / y = \tan \theta$$



Adapted from Fig. 6.1 (a) and (c), *Callister 7e*.

Poisson's Ratio

$$\nu = \frac{-\epsilon_{lat}}{\epsilon_{long}}$$

$$\nu = -\frac{(w - w_0) / w_0}{(\ell - \ell_0) / \ell_0}$$

Usually Poisson's ratio ranges from 0.25 to 0.4 and ≤ 0.5

Metals 0.25 - 0.35

Ceramics ~ 0.25

Polymers ~ 0.4

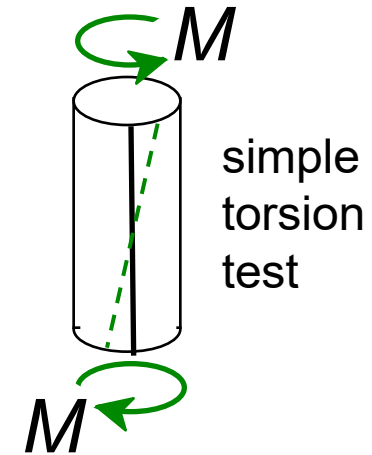
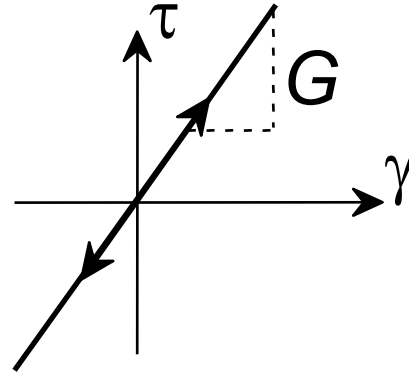
Example:

Stainless steel	0.28
Copper	0.33

Shear Modulus, G

$$\tau = G \gamma$$

τ – shear stress
 γ – shear strain

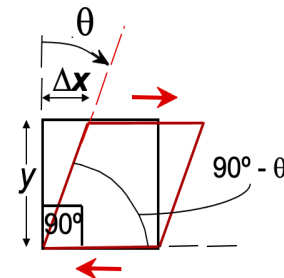


For isotropic materials (two independent parameters):

$$G = \frac{E}{2(1 + \nu)}$$

Examples

1. A cylindrical aluminum alloy test bar is subjected to a tension load. The initial diameter is 0.5 in and the initial gage length 2 in. After this cylinder is stretched from both ends, the diameter of the cylinder becomes 0.45 in and the length of the cylinder 2.5 in.
2. For the sample above what is the Poisson's ratio?
3. A block is deformed under a shear stress (i.e. parallel to a surface) as shown. If $\theta = 3^\circ$, what is the shear strain of the block?



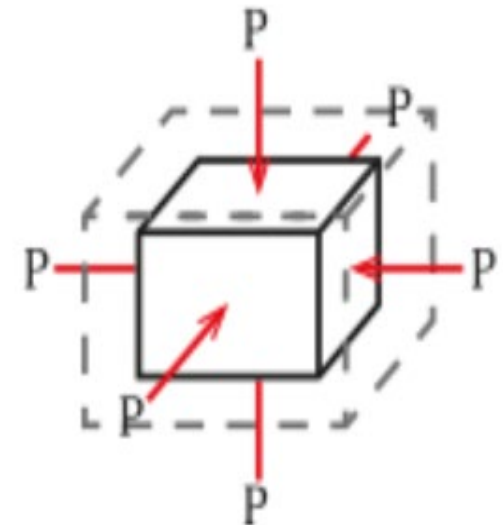
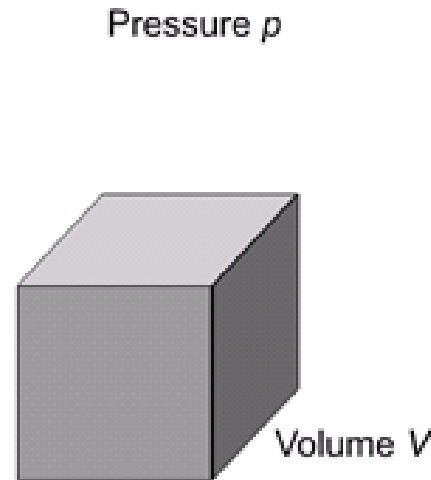
- Bulk Modulus:

$$K = \frac{E}{3(1-2\nu)}$$

*Measures resistance to bulk compression (i.e. pressure)

- Pressure:

$$P = -K \frac{\Delta V}{V_0}$$



Why is Poisson's ratio always less than 0.5?

- If $\nu = 0.5$:
$$K = \frac{E}{3(1-2\nu)} = \frac{E}{0} = \infty$$

$$P = -K \frac{\Delta V}{V_0} \quad \longrightarrow \quad \Delta V = \frac{-PV_0}{K} = \frac{-PV_0}{\infty} \sim 0 \quad \text{*Material is incompressible}$$

- If $\nu > 0.5$:
 - Under hydrostatic pressure, P , the volume would increase!
 - Unfortunately, this is not possible for engineering materials

Complete the property table for AISI 1040 Steel,
normalized at 900° C (1650° F), as below

Mechanical Properties	Value			
Modulus of Elasticity (E)	100 GPa			
Poisson's Ratio (ν)	0.30			
Bulk Modulus (K) (GPa)	(a) 38.5	(b) 76.9	(c) 83.3	(d) 167
Shear Modulus (G) (GPa)	(a) 38.5	(b) 76.9	(c) 83.3	(d) 167

True Stress-True Strain

The geometry of the sample changes to during testing.

The *True Stress* is higher than the *Engineering Stress*.

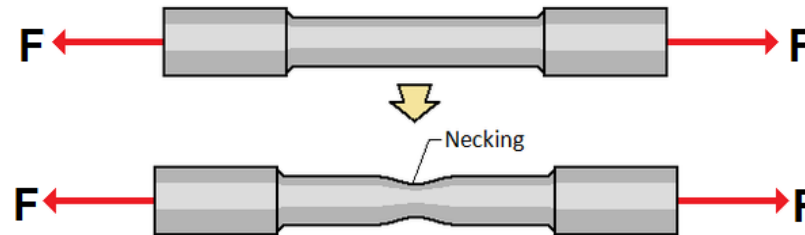
Engineering Stress

$$\sigma = \frac{P}{A_o}$$

True Stress

$$\sigma_T = \frac{P}{A_i} = \sigma(1 + \varepsilon)$$

A_i = Current Area



Engineering Strain

$$\varepsilon = \frac{\Delta l}{l_o}$$

True Strain

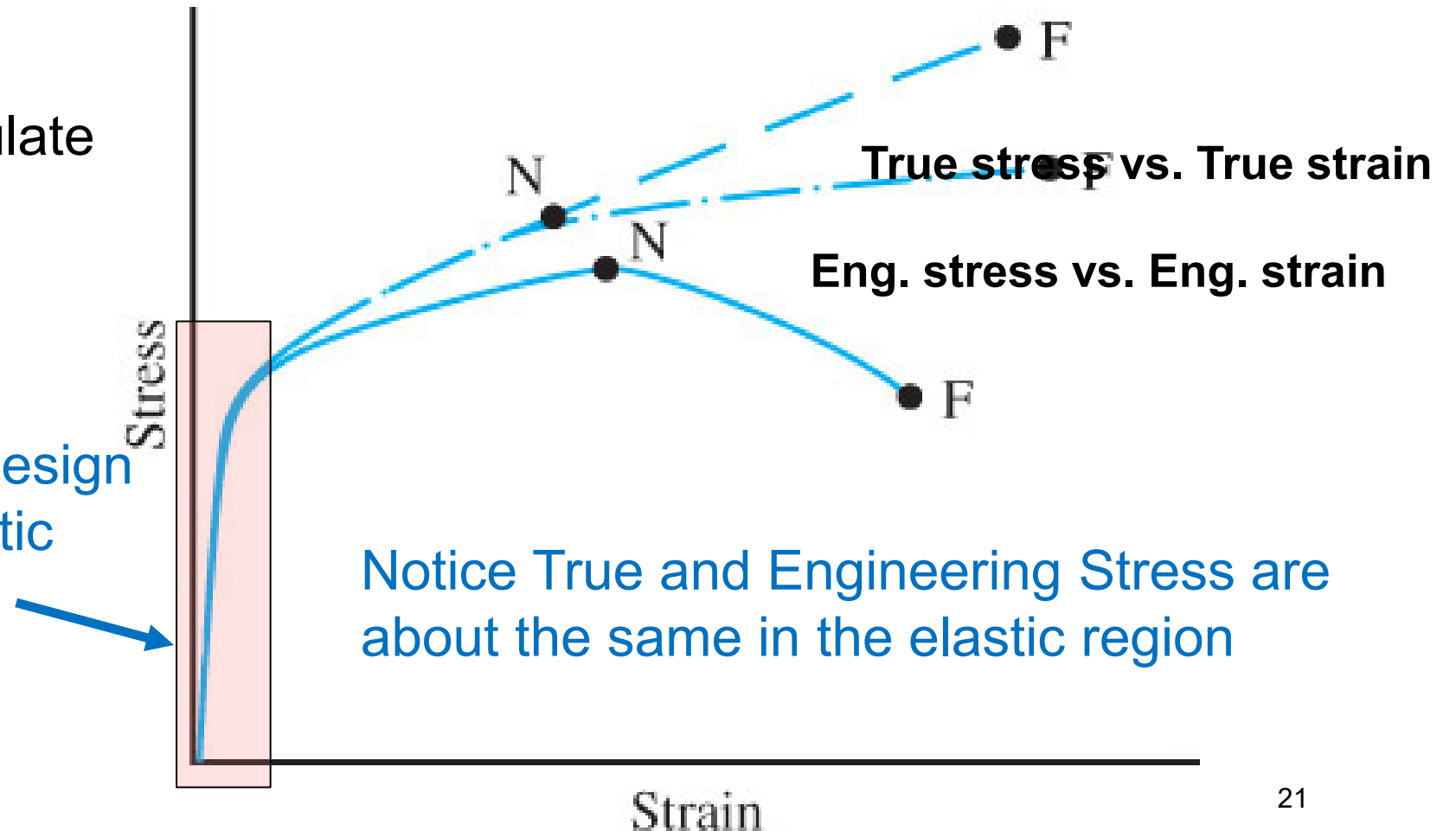
$$\varepsilon_T = \ln \left(\frac{l_i}{l_o} \right) = \ln(1 + \varepsilon)$$

l_i = Current length

Engineering stress vs. Engineering strain and True stress vs. True strain

Why do we primarily calculate engineering stress/strain?

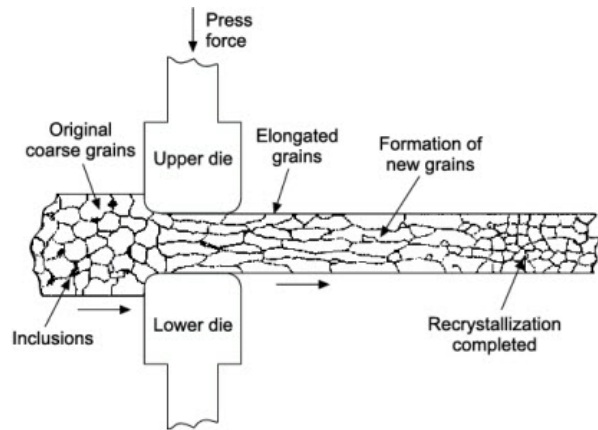
For most applications, we design for materials within the elastic range



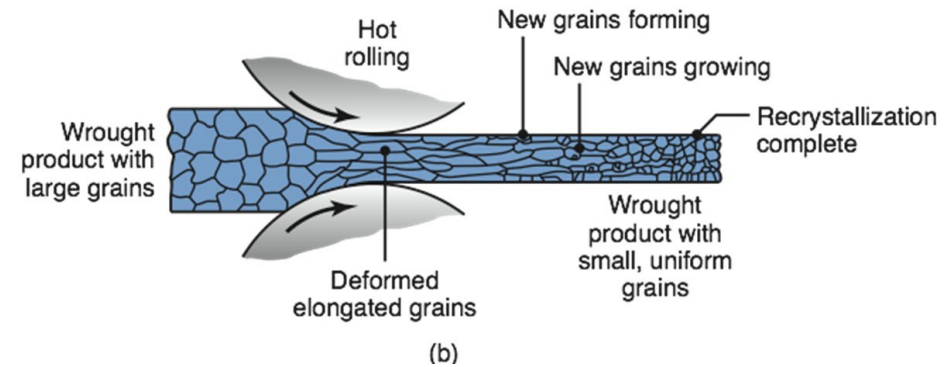
When is True Stress/Strain Used?

Certain manufacturing processes use forces to permanently shape a material:

Forging:

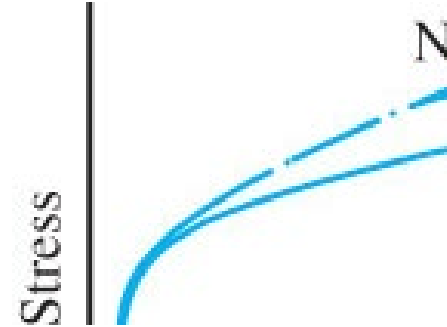


Rolling:



- For some metals and alloys from onset of plastic deformation (yielding) to the point at which necking begins (N) can be described with the following equation:

$$\sigma_T = K \epsilon_T^n$$



- Where n and K are material constants
 - n is strain-hardening exponent
 - K is not the bulk modulus here
- High n means high formability
 - i.e. a great capacity for being formed,
 - ensures that materials can be bent and stretched in a stamping die

Examples

1. An alloy test bar is subjected to a tension load. It was found that the engineering stress is 500 MPa and engineering strain 0.05. Assume no volume change occurs during deformation. The true stress and true strain in the bar are?
2. Assuming a value of 1000 MPa for K in the problem above, the strain hardening exponent n is?
3. For metal A) $n = 1.32$, for metal B) $n = 1.57$, and for metal C) $n = 1.89$, which metal has the highest formability (i.e. best for stamping) assuming all other properties and conditions are the same?